Rathete Phienes Mboki

St10522414

PDAN 8412

Part 2 Report

Table of Contents

Executive summary 3

1. Dataset choice: quality and suitability 3

2. Analysis plan 4

3. Conducted analysis: EDA and data cleaning 5

3.1 Initial inspection 5

3.2 Missing data and cleaning decisions 5

3.3 Target definition and class balance 6

3.4 Visual EDA findings 6

4. Feature selection and model training 7

4.1 Logistic regression from scratch 7

5. Model evaluation and interpretation 7

5.1 Coefficient interpretation 10

6. Retraining and addressing shortcomings 10

7. Conclusion and recommendations 10

References 11

Predicting book bestseller potential using logistic regression from scratch

Dataset: Goodreads books dataset.csv

Executive summary

The research investigates the possibility of book metadata being capable of identifying titles that can be categorized as highly rated on Goodreads. The dataset used consists of 724 titles taken from 15 fields. Since there were no official sales data or a label showing the title was on a bestseller list, it provides an educational proxy, claiming that positive books have at least 100,000 ratings. This is simply a measurement of a level of engagement, but not of the title being a bestseller from a commercial perspective. Logistic regression technique performed with a 83.5% accuracy, 78.9% precision, 51.7% recall, and 0.625 F1 score as well as the ROC-AUC value of 0.894 using the sample of 109 books kept aside from the training dataset. Although the results are still satisfactory, they are not enough for this model to be applied on a commercial level as there were not enough data used.

1. Dataset choice: quality and suitability

For the purpose of this exercise in binary classification, the dataset chosen is appropriate and correct because each row in the dataset refers to a single book, while the dataset comprises measurable variables such as average star rating, number of pages, date of publication, language, genre, description, series information, and author details. Logistic regression is an appropriate technique for calculating the probability of getting a binary result or outcome. Therefore, we can create the popularity label by analysing the rating-count value from the Goodreads database, while all other attributes can be utilized as predictors

Criterion	Evaluation and justification
Relevance	The rows describe books and contain fields plausibly associated with reader exposure and engagement.
Size and structure	724 rows and 15 columns provide enough records for a small training, validation and test exercise, although the dataset is modest for reliable generalization.
Completeness	No exact duplicate rows or duplicate IDs were found. Missingness is concentrated in series (57.60%), ISBN (23.48%), publication date (2.62%), language (2.49%), description (1.10%) and image URL (0.97%).
Quality risks	Page counts include extreme values (maximum 10,358). Ratings counts are highly skewed (median 12,735.5; maximum 9,799,364). Publication dates and author metadata may be inconsistent or missing.
Target suitability	There is no verified bestseller label or sales volume. A threshold of 100,000 Goodreads ratings creates a reproducible binary proxy: 194 positive and 530 negative books.

Leakage control	The `ratings` field defines the proxy and is deliberately excluded from model inputs. Including it would give the model the answer directly and invalidate the evaluation.
Representativeness	The sample may not represent all publishers, genres, languages, regions or newly released books. Goodreads engagement is affected by platform membership, book age and promotion.
Overall judgement	Appropriate for demonstrating a complete classification workflow, provided the report clearly labels the outcome as Goodreads popularity proxy rather than actual best seller status.

The improvement suggested here is to acquire a larger dataset which consists of validated data regarding sales figures, sales rankings, dates of publications, and features prior to publication. The advantage of doing that is that the model in question will then be able to target something commercially relevant.

2. Analysis plan

Stage	Planned steps	Output
A. EDA and data cleaning	Check the shape, column types, sample rows, duplicates, missing values, ranges, and distributions of the WRITER dataset. Pay attention to the months of publication, genres, and number of followers of authors while analysing the writer dataset.	Document data quality, clean values and understand whether the classes are imbalanced.
B. Feature engineering and selection	check for outliers and whether the dataset is balanced. classify the features of the writer dataset as publication year, genre count, length of description, series indicator, English indicator, log of number of pages, and log of number of followers of authors by omitting id numbers, URLs, and text-heavy columns like rating count.	Use interpretable predictors available from metadata and prevent target leakage.
C. Split and preprocessing	Use data partitioning of 70% for the training set, 15% for the validation set, and 15% for the testing set of the WRITER dataset. Find median, mean, and standard deviation values from the training set only and impute numeric missing values according to median training set values.	Preserve class proportions and prevent test information from leaking into preprocessing.
D. Train from scratch	Using NumPy, the algorithm of sigmoid, binary cross-entropy, gradient descent will be executed and tested with small numbers for learning rates and L2 penalty values.	Demonstrate understanding of logistic regression rather than relying on a prebuilt classifier.
E. Evaluate and interpret	The significance of this study lies in its comparison of eight different metrics, namely accuracy, precision, recall, F1, ROC-AUC, classification report, confusion matrix, and ROC curve. The authors explore the performance of their proposed model which is trained on data from adult female bandicoots.	Use complementary metrics; accuracy alone can hide missed positive cases.

F. Report and reproducibility	Briefly, describe the characteristics of the data source and target variable, cleaning methods, EDA visualizations, feature selection process, scientific approaches used for splitting of data, hyperparameter tuning, and the test results obtained together with some limitations of the analytical approach.	Make the work transparent, reproducible and understandable to the client.

3. Conducted analysis: EDA and data cleaning

3.1 Initial inspection

Everything has been completed and the CSV file is saved, which has about 724 records and includes 15 columns. There is no exact duplication of rows and there are no duplicating book IDs. The numeric data indicates that average star rating is around 3.92 while rating counts are highly skewed to the right; the median value of ratings stands at about 12,735.5 and the maximum reaches 9,799,364.

3.2 Missing data and cleaning decisions

Field	Missing values	Percentage	Decision
series	417	57.60%	Creating a binary has-series indicator missing means no series value recorded, which may not always mean a book is not in a series.
ISBN	170	23.48%	Excluded from modelling because it is an identifier and has missingness.
publication date	19	2.62%	Parse to year; invalid and missing values are imputed using the training-set median.
language	18	2.49%	Creating an English-language indicator; missing/other languages are coded 0.
description	8	1.10%	Use description length; missing descriptions become empty strings.
image URL	7	0.97%	Excluded as a non-predictive URL field.
Other modelling fields	0 reported for core fields	0%	Numeric features are checked and missing values are imputed from training data.

The document effectively parses genre and author metadata which captures incorrect values and prevents executing random strings. The number of pages and the number of author followers are log-transformed to ensure the effect of large values is minimized. Imputation of missing values and scaling of parameters is done only on the training partition.

3.3 Target definition and class balance

The goal is that the value of bestseller proxy equals 1 when Goodreads ratings reach at least 100,000, otherwise it is equal to 0. After data cleaning, the modelling table comprises 724 books: 530 of which are negatives (73.2%) and 194 are positives (26.8%). The presence of such imbalance makes one consider precision, recall, F1 score, and ROC-AUC rate along with accuracy.

3.4 Visual EDA findings

In this notebook, the following items are being generated: class balance, average ratings and log pages histograms, medians per each class, feature target correlation heatmaps, confusion matrix, ROC and training loss graphs.

Figure 1. Class balance, average star ratings and log-transformed page counts.

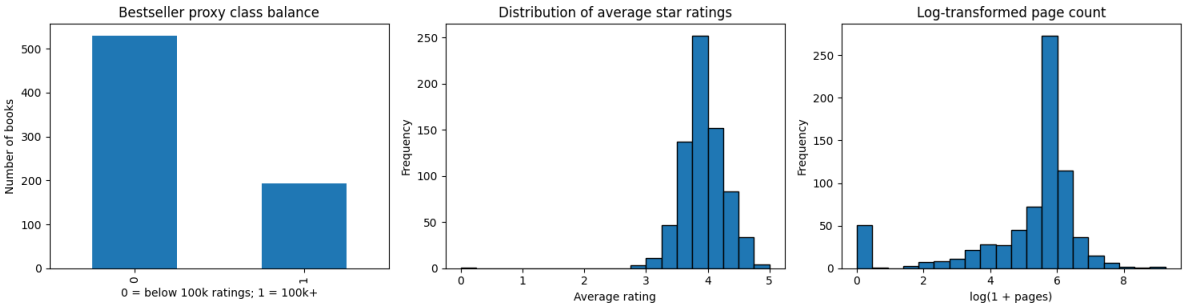
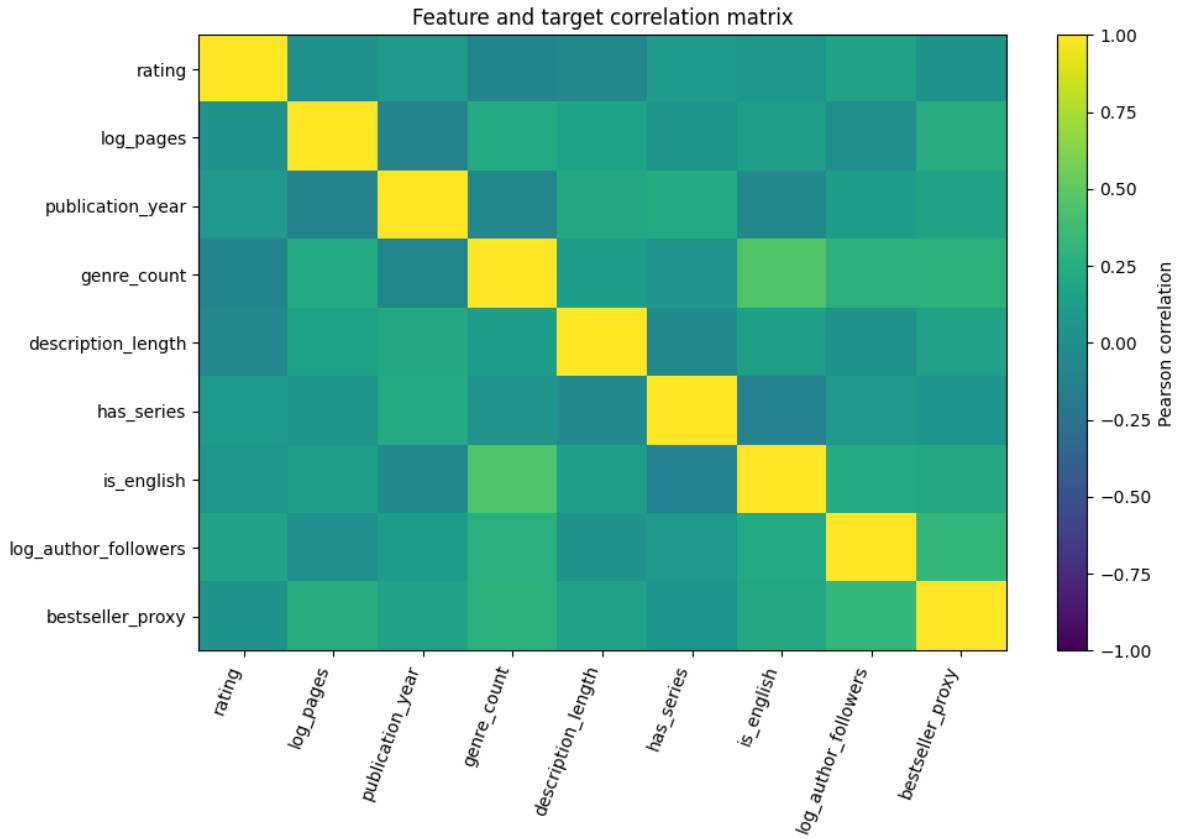


Figure 2. Correlation matrix of model features and the proxy target.



The findings on median comparison show that the books with a high level of engagement within the sample were usually published later, had longer summaries in addition to having more pages after a log transformation. The number of followers of the authors was significantly larger. When comparing the average mark, we have found out that it does not differ significantly between both groups 3.90 compared to 3.91.

4. Feature selection and model training

Eight predictors were utilized which include average rating, log page count, year of publication, genre count, length of description, series indicator, English language indicator and log author's followers. Calculation of mutual information screening was done using training data alone. According to this study, five strongest scores are log followers 0.135, log pages (0.104), description length 0.071, English indicator 0.068 and genre count 0.065. The publication year and series indicator have been given a score of zero. The dataset used was small and so it was preferred that all eight features should be considered for this purpose and future studies should work on reduced features for the future. The dataset was divided into 70% training data, 15% validation data and 15% test data. A total of 724 records were present in the final test dataset as it contained 109 books which included 80 below the threshold and 29 at or above the threshold. Missing numeric values were filled by using the median values obtained from the training data. Standardization of features was done using training means and training SDs.

4.1 Logistic regression from scratch

The algorithm calculates $z = xw + b$ and uses the sigmoid $p = 1 / (1 + \exp(-z))$ to predict the likelihood of a positive class. It minimizes binary cross-entropy with respect to L2 regularization penalty. At every iteration it makes predictions, computes weight and bias gradients, and updates variables via batch gradient descent. All the steps are implemented in the respective notebook using NumPy and comments. Three setups were compared in terms of validation F1 learning rates of 0.01, 0.03, and 0.05, all using 3,000 iterations, while L2 penalties were 0.01, 0.1, and 1.0. The best one was learning rate of 0.03, 3,000 iterations, and L2 = 0.1.

5. Model evaluation and interpretation

Metric	Test result	Interpretation
Accuracy	0.835 (83.5%)	There was a correct classification rate of test books that reached 83.5% percent. A useful number, but this does not indicate whether we missed positive books.

Precision	0.789 (78.9%)	Among the books predicted as having a high level of engagement, it was 78.9% above the proxy threshold.
Recall	0.517 (51.7%)	In fact, the model appropriated 51.7% of positive books, which makes it too conservative regarding the discovery task because more than 48.3% of positive books were missed.
F1 score	0.625	This model is characterized by balancing precision and recall, as well as having moderate results in the positive class.
ROC-AUC	0.894	This indicates that ranking is well-distinguished across the probability thresholds but does not mean that the classification result of 89.4% of correctness was reached.

Figure 3. Confusion matrix on the held-out test set.

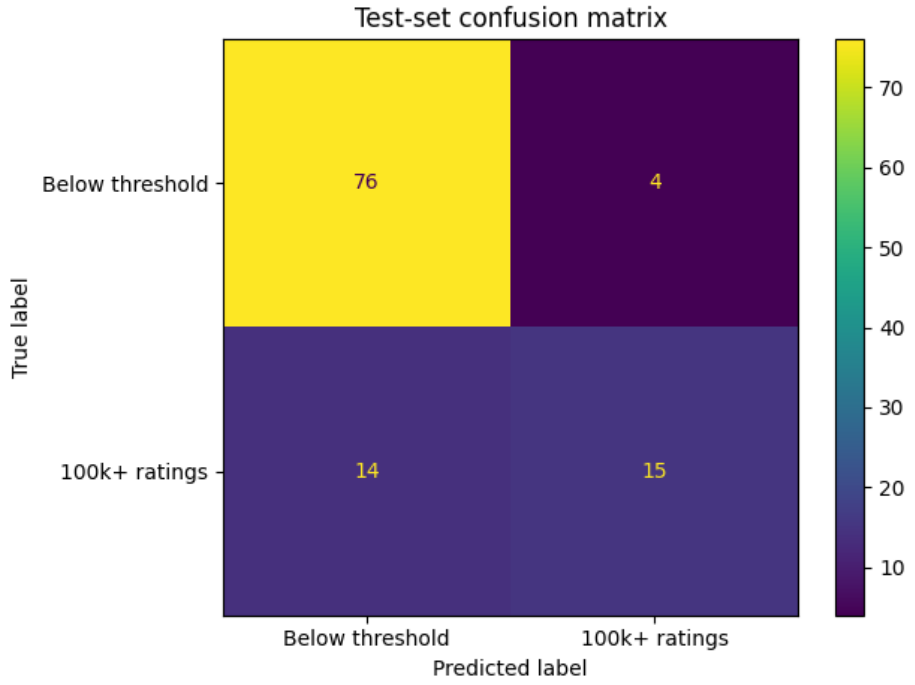


Figure 4. ROC curve on the held-out test set.

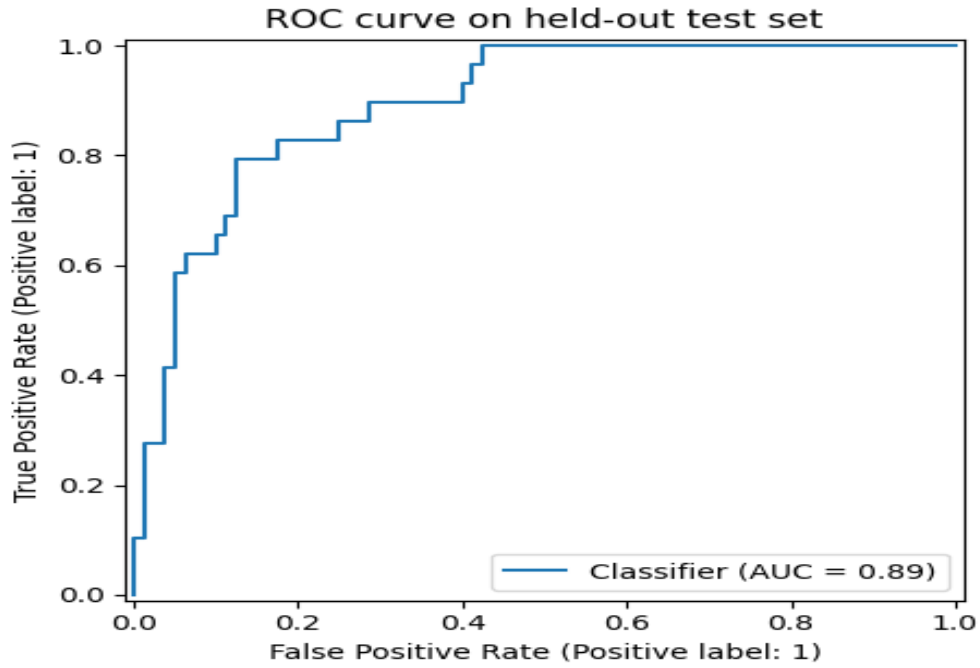
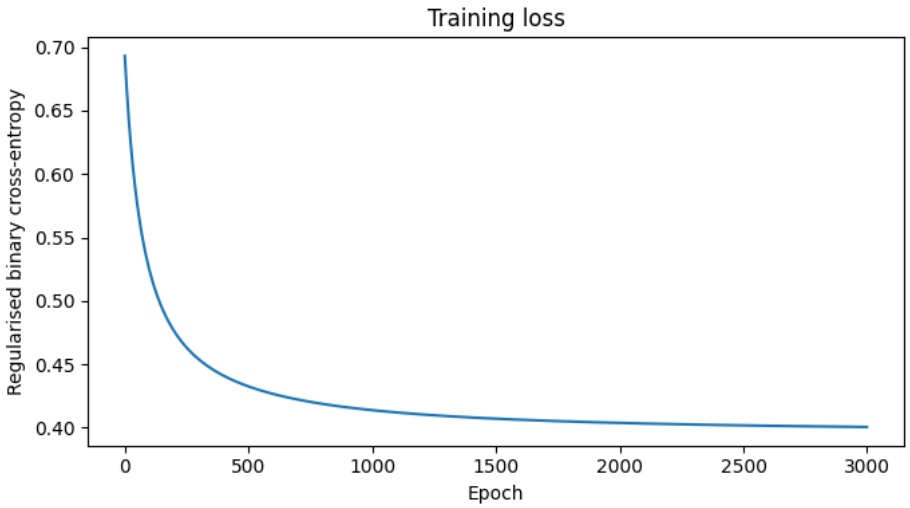


Figure 5. Training loss across gradient-descent epochs.



According to the classification report of the test, class 0 had a precision of 0.84, recall of 0.95, and F1 score of 0.89. Class 1 had a precision of 0.79, recall of 0.52 and F1 score of 0.62. In comparison with class 0, class 1 does not provide much information, but the more important thing here is the fact that class 0 has a much bigger negative class that increases the accuracy level on a general scale. This is crucial for a publisher as they need to use recall to follow hit potential candidates; in case of limited capacity of budget, precision becomes a priority. The client needs to be aware of costs – and to select the decision threshold based on this.

5.1 Coefficient interpretation

Feature	Standardized coefficient	Interpretation in this fitted model
genre count	+1.3196	Higher genre number correlates with more successful outcomes on the proxy variable, while controlling for other variables in the model.
log_author_followers	+1.1226	Higher number of author followers correlates with more successful outcomes.
log pages	+0.9811	Greater page count correlates with higher chances of success in the study sample.
is English	+0.7374	The English indicator suggests a positive relationship with the group of non-English and missing people in the combined sample.
publication year	+0.4735	The publication year also suggests a positive fitted value after adjustment.
description length	+0.1280	The length of description contains a slight positive fitted coefficient.
has series	−0.1181	The series indicator shows a slight negative fitted value.
rating	−0.0608	Average star rating has a slight negative coefficient in the multivariate model.

6. Retraining and addressing shortcomings

Three combinations of learning-rate L2 on validation data are compared in an analysis, followed up with the selection of the best of these combinations based on its validation F1 value. The selected model scores 83.5% accuracy and 0.894 ROC-AUC on the final test data, whereas its recall of a positive class is not so good at 51.7%. This is a major drawback. Proposed further experiments include Testing threshold classification probabilities among the values such as 0.35 and 0.40 on validation data, using either class-weighted loss or oversampling only on training data, making a comparison of smaller set of features and Checking results of repeated stratified or time-based cross-validation on larger data. In any case, new model should be tested on newly obtained test data.

7. Conclusion and recommendations

The project illustrates a fully functional classification process with a transparent implementation of logistic regression from scratch. Model ROC indicates that metadata has useful signals for ranking books in Goodreads. What recall shows is that by using the default 0.50 threshold the aircraft misses most of the positives. The result should be regarded as prototype data rather than as a sales forecast. Before the data can be used in practice, the press should provide real bestseller/sales info, bigger data sample, and features available now of forecasting. Once ready, the model should be constantly tested in time.

References

Goodreads dataset. (n.d.) Cleaned Goodreads books dataset supplied for this analysis. CSV file.

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) An Introduction to Statistical Learning: with Applications in R. 2nd edn. New York: Springer.

Pedregosa, F. et al. (2011) 'Scikit-learn: Machine learning in Python', Journal of Machine Learning Research, 12, pp. 2825–2830.

Note: scikit-learn is used for data splitting and evaluation metrics only. The logistic regression model itself is implemented from scratch with NumPy.